# Huawei Access Node MAR-ACC-26 QoS Congestion Event Troubleshooting SOP

**Node:** MAR-ACC-26 | **Vendor:** Huawei | **Severity:** Warning | **Alarm Code:** ALARM_5620

## Description

The Huawei Access Node MAR-ACC-26 QoS Congestion Event is a failure domain that occurs on the Huawei Access Node MAR-ACC-26 located at the Marathahalli Core site. This event is caused by Microbursts, interface tail drops, and QoS queue starvation on the Access role of the Huawei equipment. The underlying protocol mechanics responsible for this failure domain are the Huawei's implementation of the 802.1Q and 802.1p protocols, which are used for VLAN tagging and QoS prioritization. The Huawei equipment's hardware mechanics, such as the interface buffers and scheduling algorithms, also play a crucial role in this failure domain. The QoS Congestion Event is a Warning-level alarm, indicating that the QoS system is experiencing difficulties in managing the traffic flow. This can lead to packet loss, delay, and jitter, ultimately affecting the overall network performance. The Huawei Access Node MAR-ACC-26 QoS Congestion Event is a critical issue that requires immediate attention to prevent further network degradation.

## Pre-requisite Checks

- Verify that the Marathahalli Core site is accessible and the Huawei Access Node MAR-ACC-26 is powered on.

- Check the redundant link status to ensure that it is operational and not contributing to the congestion event.

- Verify that the Huawei Access Node MAR-ACC-26's firmware and software are up-to-date.

- Gather relevant logs and system information from the Huawei Access Node MAR-ACC-26 for further analysis.

## Potential Root Causes

| Cause | Probability |
|---|---|
| Microbursts on the interface causing QoS queue starvation | High |
| Interface tail drops due to high packet arrival rates | Medium |
| Incorrect QoS configuration or policy implementation | Low |

## Diagnostic Commands & Expected Outputs

```
Command: display interface brief, Expected Output: Interface status and statistics, Description: This command displays the interface status and statistics, including the number of packets received and transmitted. It helps identify if there are any issues with the interface, such as high packet arrival rates or transmission errors.
```

```
Command: display qos queue, Expected Output: QoS queue statistics and configuration, Description: This command displays the QoS queue statistics and configuration, including the queue depths and packet drops. It helps identify if there are any issues with the QoS system, such as queue starvation or packet loss.

Command: display qos policy, Expected Output: QoS policy configuration and statistics, Description: This command displays the QoS policy configuration and statistics, including the policy rules and packet matches. It helps identify if there are any issues with the QoS policy implementation, such as incorrect classification or policing.
```

## Resolution Steps

Step 1: Step: Clear interface counters, Command: clear interface, Description: This step clears the interface counters, which helps to remove any accumulated statistics and provides a clean starting point for further analysis.

Step 2: Step: Adjust QoS queue configuration, Command: undo qos queue, Description: This step adjusts the QoS queue configuration to improve queue performance and prevent starvation. This may involve adjusting the queue depths, packet drops, or other related parameters.

Step 3: Step: Review and update QoS policy, Command: undo qos policy, Description: This step reviews and updates the QoS policy to ensure it is correctly implemented and configured. This may involve adjusting the policy rules, packet matches, or other related parameters.

## Post-Check Verification

```
Command: display interface brief, Expected Output: Interface status and statistics, Description: This command verifies that the interface is stable and performing as expected, with no issues such as high packet arrival rates or transmission errors.

Command: display qos queue, Expected Output: QoS queue statistics and configuration, Description: This command verifies that the QoS queue is performing as expected, with no issues such as queue starvation or packet loss.

Command: display qos policy, Expected Output: QoS policy configuration and statistics, Description: This command verifies that the QoS policy is correctly implemented and configured, with no issues such as incorrect classification or policing.
```

## Roll-back Procedures

- Step: Restore original QoS queue configuration, Command: undo qos queue, Description: This step restores the original QoS queue configuration, which helps to revert any changes made during the resolution process.

- Step: Restore original QoS policy, Command: undo qos policy, Description: This step restores the original QoS policy, which helps to revert any changes made during the resolution process.

- Step: Reboot Huawei Access Node MAR-ACC-26, Command: reboot, Description: This step reboots the Huawei Access Node MAR-ACC-26, which helps to clear any residual issues or configurations that may have been introduced during the resolution process.